INCH-POUND

MIL-I-24739A(SH)
12 April 1991
SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

INFLATION ASSEMBLY, AUTOMATIC/MANUAL LIFE PRESERVER

Thti specification is approved for use by the Naval Sea Systems Command Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for an automatic inflation assembly with a manual override for various types of life preservers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated-Plwood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup
PPP-C-795	Cushioning Material, Packaging (Flexible Cellular, Plastic Film) for Packaging Applications
QQ-W-343	Wire, Electrical Copper (Uninsulated)

MILITARY

MILP-116	Preservation, Methods of
MIL-B-117	Bags, Sleeves and Tubing
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for
MIL-C-25369	Cylinders, Carbon Dioxide, for Inflation of Pneumatic Life Preserves

STANDARDS

FEDERAL

FED-STD-H128	Screw Thread Standards for Federal Service

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD\-167-1	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II – Internally Excited)
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields; General Requirements for

(Unless otherwise indicated, copies of federal and military specifcations and standards are available from the Standardization Documents Order Desk Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications, form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

80064-803-2482705 Life Preserver Vest MKI Main Assembly

(Copies of the drawings are available from Defense Construction Supply Center, (ATTN: DCSC-SDRL) PO Box 3990, Columbus, OH 43215.)

PUBLICATIONS

NAVSEA

S9310-AQ-SAF-010 Technical Manual for Batteries, Navy Lithium Safety Program Responsibilities and Procedures

(Copies of this publication are available from the Standardization Documents Order Desk Catalog No. 0190LP-124800, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

2.2 Non-Government publication The following documents form a part of this document to the extent specified herein unless othewise specified, the issues of these documents which are DOD adopted are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. Unless othewise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order-of-precedence In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets,

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or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample or samples shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials and components.

3.2.1 Metals. Metals shall be corrosion resistant or shall be protected to resist corrosion due to salt spray or any other environmental conditions to which this inflation assembly might be subjected during normal semice life.

322 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other unless protected against electrolytic corrosion by protective coatings.

3.23 Nonmetallic materials. All age dated nonmetallic materials used in construction of the inflation assembly shall not have been manufactured more than 24 months prior to the date of delivery except for rubber products which shall have a cure date not more than 12 months prior to. Certificates of compliance shall be provided for all age dated nonmetallic materials (see 6.3).

3.2.4 Copper shear wire The copper shear wire shall conform to QQ-W-343, type S, 22AWG, drawn and annealed

3.2.5 Rubber gaskets. Three rubber gaskets conforming to MIL-R-6855, class 2, GR. 80 shall be provided one on each end of the manifold seating areas and one for the cylinder seating area.

3.2.6 Lanyard assembly. The lanyard shall have a 100 pound tensile strength and contain a pull knob secured to the end. The lanyard shall measure 6 1/2 + 1/2 inch from the center of the D-hole (manifold mounting hole) to the end of the lanyard assembly.

3.2.7 Power supply. The power supply shall have a minimum shelf life of 3 years and be delivered with a remaining/functional shelf life of two (2) years, and be easily replaceable by the user. The installed power supply shall not react adversely by exposure to lake or seawater. Mercury or mereuric compounds shall not be used in the power supply. Power supplies containing lithium must meet the safety requirements detailed in the exemption section of ‘Technical Manual for Batteries, Navy Iithium Safety Program Responsibilities and Procedures” (S93IO-AQ-SAF-O1O) Chapter 1, paragraph l-lb, or pass all safety and performance test requirements cited in Chapter 2 of subject Technical Manual.

3.2.8 Piercing pin. Each body shall contain a piercing pin fabricated from suitable steel

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material. The piercing pin, when activated, shall pierce the puncturable area of the carbon dioxide filled cylinder, thereby releasing the carbon dioxide and permitting it to flow, without leakage, into the manifold stem and life preserver. The point of the piercing pin shall travel beyond the upper surface of the seat seal gasket and shall fully puncture the carbon dioxide filled cylinder.

3.2.9 Manual activation device. Each inflation assembly shall contain a lever or mechanical device, fabricated from suitable material, for manual activation. The device shall be secured by copper shear wire. In the unoperated position, if a lever is used, the lever shall lay against the side of the body and shall not exceed two (2) inches. The design of the device shall be compatible for use with a manifold conforming to drawing 803-2482705. A single strand of soft, uncoated, copper wire, conforming to QQ-W-343, type S shall secure the lever to the body

3.2.10 Body. The body shall be fabricated from a high impact, shock resistant material and shall be free from blow-holes, porosity, cracks, or other defects. Each body shall have internal threads 1/2-20 UNF-2B conforming to FED-STD-H28 for use with carbon dioxide cylinders conforming to MIL-C-25369, type II. The opening in the body shall contain a chamfered portion and shall be so designed to permit the top area of the carbon dioxide (CO2) filled cylinder to seat firmly against the seat of the body. The seat shall be smooth, flat and perpendicular to the threads. The body shall contain a gas passage and a means to mount assembly on the manifold stem conforming to NAVSEA Drawing 803-2482705.

3.2.11 Recovered materials Unless othewise specified herein, all equipment, material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 General description. The automatic/manual inflation assembly shall be an actuation mechanism designed to operate automatically and manually when properly attached to inflation manifold conforming to NAVSEA drawing No. 803-2482705. It shall be designed to accept a CO2 Cylinder conforming to MIL-C-25369, type IL The manual activation device shall be actuated by a lanyard and knob assembly.

3.4 Design. The inflation assembly shalI consist of a body with a seat seal gasket for use with a carbon dioxide cylinder, an outlet with gas passage, a manual activation device with a lanyard a piercing pin, and a water activated initiator device. The inflation assembly shall automatically activate within 3 seconds when fully immersed in lake or seawater having a conductivity of 100 +50/- 10 microsiemens (uS) and higher which causes the piercing pin to move forward, puncturing the carbon dioxide filled cylinder, releasing the gas, and thereby, inflating the preserver. Each inflation assembly shall complete ten (10) consecutive manual operation. After each manual

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operation the procedure for resetting the assembly shall be limited to returning the manual activation device to the original position, replacing the shear wire, and installing a new CO2 cylinder. The capability for an inflation assembly with multiple automatic operations is acceptable only if the reset procedures do not require any disassembly of the inflation assembly and has a positive means of verification of resetting. Assemblies that do not have multiple automatic operation capability shall be so designed that after automatic operation the assembly is unable m be reset to prevent the possibility of deployment of an assembly which no longer has automatic operating capability.

3.4.1 Dimensions.

Length - of the assembly shall be measured along an axis that is parallel to the flat side of the manifold D-hole.

- length including the CO2 cylinder, when measured from the center of the D-hole shall not exceed 7 inches.

- length from center of manifold D-hole to either edge shall not exceed 2 inches.

Width – from center of inflation manifold D-hole to either edge shall not exceed 1 1/4 inches.

Height – shall not exceed 1 1/8 inch.

3.4.2 Weight The inflation assembly without CO2 cylinder shall weigh no more than 11 ounces (ozs) (311.8 Grams).

3.5 Performance.

3.5.1 Piercing pin. When inspected as specified the piercing pin shall not puncture a filled CO2 cylinder when fully installed by hand (see 4.6.1).

3.5.2 Manual operation with the carbon dioxide filled cylinder. For this test the inflation assembly shall be installed on an inflation manifold conforming to 803-2482705. The piercing pin of the inflation assembly shall advance to its maximum point of travel and shall pierce the puncturable area of an installed carbon dioxide filled cylinder conforming to MIL-C-25369, type II, when inspected as specified with a maximum load of 15 pounds attached to the free end of the lanyard. The CO2 cylinder shall be My discharged within ten (10) seconds. The gas shall not pass through any part of the inflation assembly into the atmosphere other than through the gas passages in the outlet. The carbon dioxide filled cylinder, when fully inserted with hand tightening, may make contact with the point of the piercing pin, but the cylinder shall not be punctured. The inflation assembly shall then conform to the requirements for the automatic operation of 3.5.4 (see 4.6.2).

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3.53 Repeated manual operation with the carbon dioxide filled cylinder. The inflation assembly shall be capable of completing ten (10) consecutive manual operations as described in 3.5.2, followed by one (1) automatic actuation conforming to the requirements of 3.5.4. It is acceptable to warm inflation assembly back up to room temperature between activations (see 4.6.3).

3.5.4 Automatic operation. When inspected as specified, the piercing pin of the inflation assembly shall advance to its maximum point of travel and shall pierce the puncturable area of an installed carbon dioxide filled cylinder conforming to MIL-C-25369, type II, in a maximum time of 3.0 seconds The C02 cylinder shall be fully discharged within a total of ten (10) seconds. The gas shall pass through the gas passages in the outlet into the life preserver carbon dioxide compartment and shall not pass through any part of the inflation assembly into the atmosphere. The products resulting from the firing of the CO2 cylinder shall not cause damage to the life preserver. The carbon dioxide filled cyinder, when fully inserted with hand tightening, may make contact with the point of the piercing pin but shall not puncture the cylinder (see 4.6.4).

3.6 Support equipment There shall be no special or non-standard equipment required for the installation, operation, reset capability, or maintenance of the inflation assembly.

3.7 Vibration and explosive atmosphere.

3.7.1 Vibration. When tested and inspected as specified, the inflation device shall not be adversely affected or function inadvertently when exposed to vibration conditions (see 4.9).

3.7.2 Explosive atmosphere. After completion of the vibration test, the same inflation device shall be tested and inspected as specified in paragraph 4.9.2. The inflation device shall operate in flammable atmosphere without causing an explosion.

3.8 Rain. When tested and inspected as specified, the inflation device shall not be adversely affected or function inadvertently when exposed to rain conditions (see 4.10).

3.9 Salt fog. When tested and inspected as specified, the inflation device shall not be adversely affected or function inadvertently when exposed to salt fog (see 4.11).

3.10 Temperature When tested and inspected as specified, the inflation device shall not be adversely affected or function inadvertently by exposure to high and low temperatures (see 4.7.1 and 4.7.2).

3.11 Humidity. The device shall not be adversely affected or function inadvertently when subjected to the renditions of humidity resistance (see 4.12).

3.12 Hydrostatic pressure The device shall not be adversely affected or function

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inadvertently when subjected to a hydrostatic pressure (see 4.13).

3.13 Differential pressure. The device shall not be adversely affected or function inadvertently when subjected to surge air pressure (see 4.14).

3.14 Electromagnetic radiation. The device shall not be adversely affected or function inadvertently when subjected to an electromagnetic environment (see 4. 15).

3.15 Impact. The device shall not be adversely affected or function inadvertently when subjected to a free fall and impact (see 4.16).

3.16 Visual. The inflation assemblies shall be of uniform quality, and free from other defects which could adversely affect performance, reliability or durability (see 4.5).

3.17 Marking. Each inflation assembly shall have a permanently affixed label with legible and durable letters and numerals in accordance with MXLSI'D-130 with the following information:

Inflation Assy., Automatic
Contract #_____
Mfgr____(Namel_____
SER #_____
Lot #_____
NSN___(To be assigned]
Date of Mfgr_____

Battery/replacement/dates



3.18 Serial Number. Each item shall be identified by individual serial numbers which shall be assigned by the manufacturer (see 6.2). Serialization shall be by a block of consecutive numbers to cover the entire acquisition document quality. Method of marking shall be as specified in 3.18.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of aIl inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance Each inflation assembly shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not releive the contractor of the responsibility of ensuring

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that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does It commit the Govemment to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein arc classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. Except where otherwise specified 15 inflation assemblies shall be examined and tested by a Government approved Commercial laboratory or a Government laboratory as specified by the contracting officer. All 15 of the inflation assemblies shall be tested and examined in accordance with the examination list in table L Each inflation assembly shall then be tested and examined in accordance with a different one of the remaining first article inspections as specified in table I. If one (1) of the sample inflation assemblies fails any test specified in table I, the assemblies represented for first article inspection shall be rejected (see 6.4). When a contractor is in continuous production of these automatic inflation assemblies from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or waiving the first article inspection does not exempt the manufacturer from the requirements for performing the quality conformance inspection-

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TABLE I. *Inspections.*

Inspection	Requirement	Test	First article inspection	Quality conformance inspection
Visual examination	3.16	4.5	x	x
Vibration	3.7.1	4.8.1	x	
Explosive atmosphere	3.7.2	4.8.2	x	
Hydrostatic pressure	3.12	4.12	x	
Impact	3.15	4.15	x	x
salt fog	3.9	4,10	x	x (1 unit)
Differential pressure	3.13	4.13	x	
Electromagnectic radiation	3.14	4.14	x	
Rain	3.8	4.9	x	
Temperature	3.10	4.7 4.7.1 4.72	x x	X (1 unit) x (1 unit)
Humidity	3.11	4.11	x	
Dimensions and weight	3.4.1 3.4.2	4.18	x	x
Piercing pin	35.1	4.6.1	x	x
Manual operation w/cylinder	3.5.2	4.6.2	x	x
Repeated manual operation w/cylinder	353	4.63	x	x (1 unit)
Auto operation w/cyiinder	3.5.4	4.6.4	x	x

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4.4 Quality conformance inspection. Except where noted, the sample inflation assemblies selected in accordance with 4.4.2 shall be subjected to all of the quality conformance inspections as specfied in table I. Any sample which fails any quality conformance inspection shall be rejected. The four (4) tests in table I which are annotated with "1 unit" shall require only one (1) sample inflation assembly subjected to the test requirement.

4.4.1 inspection lot For the purpose of quality conformance inspection, a lot is defined as all inflation assemblies of one type, produced in the same facility under essentially the same conditions, using the same batches of materials.

4.4.2 sampling for quality conformance. As a minimum, the contractor shall select a sample quantity of inflation assemblies in accordance with table II and subject them to the quality conformance inspections and tests listed in table I. Sample size depends on the classification of the characteristic as show in table III. If one or more defects is found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100% of the rejected lot for the defective characteristic or providing a new lot which shall be inspected in accordance with the sampling plan defined herein.

TABLE II. *Sampling for quality conformance.*

lot *Size*	Sample size	
	Major defects	Minor defects
2 to 50	5	3
51 to 90	7	6
91 to 150	11	7
151 to 280	13	10
281 to 500	16	11
501 to 1,200	19	15
1,201 to 3,200	23	18
3,201 to 10,000	29	22
10,001 to 35,000	35	29
35,001 to 150,000	40	29

4.5 Visual examination. Each sample inflation assembly selected in accordance with 4.4.2 shall be visually, examined to determine conformance to this specification. The sample shall be examined in accordance with table I, for defects as specified in table III. Any sample failing to conform to table III examination requirements shall be rejected

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TABLE III *Classification of defects.*

Categories	Defects
Major:	
101	Any cracks
102	Any voids
103	Any missing or hose parts
104	Any burrs or sharp edges
105	Any improperly assembled components
106	Any improper materials
107	Any corrosion
108	Fails operation test(s)
109	Fails enviromental test(s)
Minor:	
201	Any foreign matter
202	Any flaked surface coating
203	Any missing, incorrect or illegible markings
204	Length lanyard not as required
205	Any frayed lanyard ends
206	Any dents
207	Dimensions and weight not as specified

4.6 Performance.

4.6.1 Piercing pin. A carbon dioxide filled cylinder conforming to MIL-C-25369 type II shall be fully installed in the inflation assembly by hand tightening The assembly shall then be inspected to ensure the cylinder has not been punctured by listening for the sound of escaping gas. The carbon dioxide filled cyiinder shell then be removed and the top of the cyiinder examined for evidence of puncture and leakage (see 3.5.1).

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4.6.2 Manual operation with the carbon dioxide filled cylinder. The manual activation device shall not be secured to the body with copper shear wire during this inspection. A carbon dioxide filled cylinder conforming to MlL-C-25369, type II, shall be fully installed in the inflation assembly with hand tightening. The inflation assembly shall be examined to ensure that the top of the cylinder has not been punctured with the point of the piercing pin by listening for the sound of escaping gas. The carbon dioxide filled cylinder shall then be removed and the top of the cylinder examined. Upon completion of the examination, the carbon dioxide cylinder shall be fully reinstalled in the inflation assembly with hand tightening. The inflation assembly shall be held in a vertical position with the cylinder at the top. The device shall be in its normal position and a maximum load of 15 pounds shall be attached to the free end of the device. The load shall operate the device through one normal cycle. The piercing pin shall advance to its maximum point of travel and shall puncture the carbon dioxide filled cylinder. The CO2 cylinder shall be fully discharged within seven (7) seconds after activation. The gas shall not pass through any part of the inflation assembly into the atmosphere other than through the gas passages in the outlet. The empty cylinder shall be removed and the top examined for the location of the puncture (see 3.5.2).

4.6.3 Repeated manual operation with the carbon dioxide filled cylinder. The manual activation device shall not be secured to the body with copper shear wire during this inspection. A carbon dioxide filled cylinder conforming to hKILC-25369, type II, shall be fully installed in the inflation assembly with hand tightening. The inflation assembly shall be examined to ensure that the top of the cylinder has not been punctured with the point of the piercing pin by listening for the sound of escaping gas. The carbon dioxide filled cylinder shall then be removed and the top of the cylinder examined. Upon completion of the examination, the carbon dioxide cylinder shall be fully reinstalled in the inflation assembly with hand tightening. The inflation assembly shall be held in a vertical position with the cylinder at the top. The device shall be in its normal position and a maximum load of 15 pounds shall be attached to the free end of the device The load shall operate the device through one normal cycle The piercing pin shall advance to its maximum point of travel and shall puncture the carbon dioxide filled cylinder. The CO2 cylinder shall be fully discharged within seven (7) seconds after activation. The gas shall not pass through any part of the inflation assembly into the atmosphere other than through the gas passages in the outlet. The empty cylinder shall be removed and the top examined for the location of the punctured Each inflation device shall be capable of completing ten (10) consecutive manual operations (see 3.5.3).

4.6.4 Automatic operation and examination. A carbon dioxide filled cylinder conforming to MIL-C-25369, type II, shall be fully installed in the inflation assembly with hand tightening. The inflation assembly shall be immersed in water with a conductivity of 100 +50/-10 microsiemens and temperature of 70 + 3 degrees Fahrenheit. The CO_2 cylinder shall be punctured in a maximum time of 3.0 seconds, the time being measured from the instant the inflation assembly is fully immersed The test shall be terminated if 3.0 seconds is exceeded. The piercing pin shall advance to its maximum point of travel and shall puncture the C02 filled cylinder. The CO2 cylinder shall be fully discharged within ten (10) seconds after immersion. The gas shall be immediately released and the piercing pin shall not obstruct or retard the flow of the gas. The gas shall not pass through any part of the inflation assembly other than the gas passages. The body of

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the inflation assembly shall be examined and shall not have shattered *or* fragmented. The empty cylinder shall be removed and the top examined for the location of the puncture (see 3.5.4).

4.7 Temperature exposure.

4.7.1 High temperature. The inflation assembly without the copper shear wire shall be subjected to a temperature of 160 + 5 degrees Fahrenheit (71 + 3 degrees Centigrade) in accordance with MIL-STD-810 method 501.2, Procedure 1 (storage), for 48 + 1/2 hours. The inflation assembly shall then be immediately removed and immersed in water having a temperature of 60 + 3 degrees Fahrenheit (15 + 2 degrees Centigrade) and a conductivity of 100+50/– 10 microsiemens. The assembly shall then conform to the requirements of paragraph 3.5.4. After a minimum of' 1/2 hour under water the inflation assembly shall be removed and examined. There shall be no evidence of cracks, flaking, deformation, distortion, splintering, fragmentation or other defects (see 3.10).

4.7.2 Low temperature. The inflation assembly with the copper shear wire installed shall be subjected to a temperature of minus 20 + 2 degrees Fahrenheit (minus 28 + 1 degrees Centigrade) for a minimum of 48 + 1/2 hours in accordance with MIL-STD-810 method 5022, Procedure I, (storage). The inflation assembly shall then be immediately removed and immersed in water having a temperature of 32 + 2 degrees Fahrenheit (zero +/ degree Centigrade) and a conductivity of 100 +50/-10 microsiemens The inflation assembly shall conform to the requirements of paragraph 3.5.4. After being immersed for a minimum of 1/2 hour, the inflation assembly shall be removed from the water and examined. There shall be no evidence of cracks, flaking, deformation, distortion splintering, fragmentation, or other flaws nor shall the assembly be activated The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.10).

4.8 Vibration and explosive atmosphere.

4.8.1 Vibration. The inflation assembly shall be subjected to vibration conditions in accordance with MIL-STD-167-1, type I. After testing, the assembly shall show no evidence of nicks, cracks, loose particles, or other flaws, nor shall the assembly be activated. The assembly shall then conform to the requirements of 4.9.2.

4.8.2 Explosive atmosphere The inflation assembly shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, Method 511.2, Procedure I, with test altitude being sea level and ambient air temperature of 125 + 5 degree Fahrenheit (52 + 3 degrees centigrade.)

NOTE No identification plate reflecting testing information is required to be attached to the inflator after testing is completed.

4.9 Rain. The inflation assembly shall be subjected to the rain water test in accordance with MIL-STD-810, Method 506.2, Procedure II, with rain water having a conductivity of 350 + 35

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microsiemens with a fall rate of four (4) inches per hour. After testing, the assembly shall show no evidence of nicks, cracks, loose particles, or other defects, nor shall the assembly be activated. The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.8).

4.10 Salt fog. The inflation assembly shall be subjected to the conditions of salt fog test in accordance with MIL-STD-81 O, Method 509.2, Procedure I. Test duration is to be for a constant 48 hours salt fog and then 48 hours drying. After testing, the assembly shall show no evidence of nicks, cracks, loose particles, or other defects, nor shall the assembly be activated. The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.11).

4.11 Humidity. The inflation assembly shall be subjected to the renditions of the humidity resistance test in accordance with MIL-STD-810 Method 507.2, Procedure III using the temperature/humidity cycle from figure 507-2-3 of the test method. After testing, the assembly shall show no evidence of cracks, flaking, loose particles, deformation, distortion or any other defects, nor shall the assembly be activated. The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.11).

4.12 Hydrostatic pressure A discharged carbon dioxide cylinder shall be inserted in the inflation assembly with a maximum torque of 20 inch pounds and the copper shear wire shall be installed. The assembly shall be subjected to a hydrostatic pressure of 1500 pounds per square inch (lb/in2) applied to the outlet of the inflation assembly for a minimum of 5 minutes. The inflation assembly shall then be inspected and there shall be no evidence of leakage or deformation of the body. After testing the assembly shall conform to the requirements of 3.5.2 and 3.5.4 (see 3.12).

4.13 Differential pressure The inflation assembly shall have the outlet hole for escape of pressurized gas blocked off fix this test The inflation assembly, equipped with a filled carbon dioxide cylinder, shall be positioned in a test chamber and subjected to three cycles of surge air pressure, of 40 + 2 lb/in2 for a duration of 50 + 20 milliseconds, to simulate blast conditions. The assembly shall not show any evidence of damage or function inadvertently. After testing the assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.13).

4.14 Electromagnetic radiation. The inflation assembly shall be subjected to electromagnetic environment level in accordance with MKL-STD-1385, table I. After testing the assembly shall show no evidence of cracks, deformation, distortion, or any other flaws, nor shall the assembly be activated. The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.14).

4.15 Impact The inflation assembly shall be subjected to the following test. A filled carbon dioxide cylinder shall be fully installed in the inflation assembly. The assembly shall be free fall dropped from a minimum height of 4 feet onto a fixed steel plate, twice on each face. After testing, there shall be no evidence of cracks, deformation distortion, splintering fragmenting or any other flaws, nor shall the assembly be activated. The assembly shall then conform to the requirements of 3.5.2 and 3.5.4 (see 3.15).

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NOTE Minor deformation or distortion is acceptable provided the assembly still conforms to all stated requirements.

4.16 Workmanship. The inflation assemblies shall be inspected to ensure the following: Uniform quality, clean, smooth, free from waviness and bubbles, splits, cracks, nicks, burns, sharp edges, deformation, distortion, or any other irregularities or defects which could adversely affect the performance reliability or durability of the inflation assembly (see 3.16).

4.17 Packaging inspection. The preservation, packing, and marking shall be inspected to verify conformance to the requirements of section 5.

4.18 Dimensions and weight.. When inspected in accordance with table I for the defects as specified in table III, the inflation assembly shall meet the requirements of 3.4.1 and 3.4.2.

5. PACKAGING

5.1 Preservation, Preservation shall be Level A or Commercial (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning The inflation assembly shall be cleaned in accordance with MIL-P-1 16, Process C-1.

5.1.1.2 Preservative application. Preservatives shall not be used.

5.1.1.3 Unit packs. Each inflation assembly shall be individually unit packed in accordance with sub-method 1A-8 of MIL-P-116 in a bag conforming to MIL-B-1 17, type I, Class E, Style 2. Each bag shall be fabricated from a continuous piece of barrier material. Cushioning shall conform to PPP-C-795, class 3. Each unit shall be placed in a supplementary container to variety 2 of PPP-B-566 or PPP-B-676.

5.1.1.4 Intermediate packs. Unit pack 10 each shall be placed in intermediate container conforming to PPP-B-636 Class weather-res istant. Intermediate containers shall be uniform in size and shape.

5.1.1.5 Commercial. Each inflator assembly shall be individually packaged in a waterproof container and meet the requirements of ASTM D3951.

5.2 Packing. Packing shall be level A B, C, or commercial, as specified (see 6.2).

5.2.1 Level A Inflation assemblies, preserved as specified in 5.1, shall be packed in wood containers conforming to PPP-B-601, overseas type or PPP-B-621 class 2. Closure shalI be in accordance with the applicable container specification.

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5.2.2 Level B. Inflation assemblies, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, Class weather-resistant, style optional. Closure, waterproofing and reinforcing shall be in accordance with the container specification.

5.23 Level C. Inflation assemblies, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, Class Domestic, Style optional. Closure shall be in accordance with the applicable container specification.

5.2.4 Commercial. Ten inflator assemblies shall be packed in accordance with ASTM D3951.

5.3 Marking.

5.3.1 Standard marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit, supplementary, intermediate, and exterior container shall be marked in accordance with MIL-STD-129. The complete military or contractors type or part number, as applicable (including the Commercial and Government Entity Code (CAGE Code), shall be marked on all units, supplementary and intermediate packs in accordance with the identification marking provisions of MIL-STD-129.

5.3.2 Special marking. As specified for hazardous materials in MIL-STD-129, Appendix E.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use The Automatic/Manual Inflation Assembly covered by this specification is intended for use on various life preservers used aboard ships. It is not for use by aircrew.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. When first article is required (see 3.1 and 4.3)

d. National stock number and quantity

e. Serial numbers to be assigned by the manufacturer (see 3.20)

f. First article inspection may be waived (see 43)

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g. Level of applicable preservation, and packing requirements for special markings (see 5.1, 5.2, and 5.3)

h. First article inspection shall. be performed by a Government approved commercial laboratory or a Government laboratory as specified by the contracting officer (see 4.1)

i. Nonmetallic materials certification (see 3.2.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contact. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To insure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supple-merit 27.475-1 exempts the requirements for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.3	DI-E-2121	Certificate of Compliance	

The above DID's were those cleared as of the data of this specifi-cation. The current issue of DoD 501 O.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractors current inventory and the number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1 and 4.3).

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6.5 Subject term (key word) listing.

Cylinder
Lanyard
Microsiemens

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy – SH
(Project 4220-N376)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-24739A(SH)	12 April 1991

3. DOCUMENT TITLE

Inflation Assembly, Automatic/Manual, Life Preserver

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander Naval Sea Systems Command (SEA 55X24)	b. TELEPHONE (Include Area Code) (1) Commercial (703) 602-0323	(2) AUTOVON AV-332-0323
c. ADDRESS (Include Zip Code) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office	

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